

Changdao He

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Education

University of Alberta, MSc in Computing Science Sep 2026 – Present

- Research interest: Online Algorithms
- Advised by Professor Xiaoqi Tan

University of Toronto, BSc in Computer Science, Mathematics Sep 2022 – Apr 2026

- Graduated with High Distinction
- GPA: 3.80/4.0

Research Experience

Studies in Online Throughput Scheduling Problems Jul 2025 - Present

Supervised by Prof. Allan Borodin, University of Toronto

- Developed deterministic online algorithms and impossibility results for real-time throughput scheduling under multiple preemption models, including preemption–restart and preemption–revoke; analyzed competitive ratios through adversarial constructions and algorithm design
- Resulting papers:
 - Revisiting Real-Time Interval and Throughput Maximization (with Allan Borodin and Nadim Mottu). *arXiv preprint arXiv: 2607.16163*
 - Revoke vs. Restart in Unweighted Throughput Scheduling. *arXiv preprint arXiv: 2510.15318*

Automated Puzzle Solving, Generation, and Difficulty Estimation Sep 2025 - Present

Co-supervised by Prof. Jonathan Calver and Prof. Alice Gao, University of Toronto

- Exploring computational approaches to logic puzzles, with three main goals: implementing solvers, generating new puzzles, and categorizing puzzle difficulty
- Investigating SAT-based formulations of puzzle-solving, and comparing solver-based vs. human difficulty perspectives
- Studying methods for puzzle generation and evaluation to enable difficulty-controlled puzzle creation

Monte Carlo Tree Search for Othello May 2025 - Aug 2025

Supervised by Prof. Alice Gao, University of Toronto

- Implemented a Monte Carlo Tree Search agent for Othello, focusing on randomized simulations for decision-making in complex states
- Compared MCTS against minimax-based approaches and integrated heuristic playouts to improve simulation quality

Search and Heuristic Methods for Othello Jan 2025 - Apr 2025

Supervised by Prof. Alice Gao, University of Toronto

- Implemented an AI agent for Othello using minimax search with alpha–beta pruning
- Designed and tested multiple heuristic board-evaluation functions to enhance performance
- Analyzed trade-offs between search depth, pruning efficiency, and heuristic accuracy

Presentations

- "Monte Carlo Tree Search for Othello", Summer Research Poster Showcase, Department of Computer Science, University of Toronto (Aug 2025) (*Poster available at changdaohe.com*)

Internship Experience

Software Development Intern, BGI Genomics – Shenzhen, China Jul 2023 – Aug 2023

- Enhanced CRM system modules, performed functionality testing, and explored automation strategies to streamline workflows
- Organized and led group meetings, demonstrating strong coordination and project management skills

Data Processing Intern, Shenzhen University – Shenzhen, China Jun 2023

- Processed and visualized EEG data using Python (matplotlib)
- Collaborated with graduate students to determine effective data visualization strategies

Honours & Awards

- **Dean's List Scholar**, Faculty of Arts & Science, University of Toronto (2023, 2024, 2025, 2026)